(In)direct evidential futures in Colloquial Jakartan Indonesian*

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1. Introduction

In many languages, speakers can convey how they came to know the information they are stating, often through grammatical devices associated with evidentiality (cf. Aikhenvald 2004). Evidential markers may appear as affixes or as separate words, and languages with evidentials differ in whether their use is obligatory. The kinds of evidential contrasts encoded vary widely: some languages mark distinctions like visual versus non-visual perception, while others focus on the speaker's level of certainty or the means by which they acquired the information. A particularly prevalent contrast is between *direct* and *indirect* evidence—commonly glossed as *firsthand* versus *non-firsthand*—which signals whether the speaker personally witnessed the event or is reporting it based on inference, hearsay, or other indirect means. To illustrate this distinction, consider Example (1) from Shipibo-Konibo below. In Example (1a) a reportative evidential is used, since the information is acquired through a book, and thus not directly. By contrast, in Example (1b) a direct evidential is used, since the information is acquired directly, through visual perception.

(1) a. Alemania-**ronki** Holanda patax iki
Germany:ABS-REP the.Netherlands next.to COP
'Germany is next to the Netherlands (I read it in a book).'

b. Alemania-**ra** Holanda patax iki.
Germany:ABS-DIR.EV the.Netherlands next.to COP
"Germany is next to the Netherlands (I saw it on a map).'

Shipibo-Konibo, Valenzuela (2003):ex.46

*I thank Amir Anvari, Amy Rose Deal, Kai von Fintel, Martin Hackl, Sabine Iatridou, David Pesetsky, Viola Schmitt, the instructors and participants of the Fieldwork and the Workshop classes at MIT, the audiences at SuB, NELS, AFLA, PLC, UCSD, LFRG, and SNEWS. Thanks to Tia Wulansari for sharing with me her language.

Cross-linguistically, there seems to be a typological gap: there are no *direct* evidentials in the future (Visser 2015). This is in a sense expected, since the future has not yet occurred, and thus cannot be directly perceived. But linguistic categories do not need to be constrained by our naive metaphysics. Even if directly experiencing the future is impossible, this does not mean that we cannot have a way of talking as if we can. In other words, it could be that for the purposes of language there is a notion of what counts as 'direct enough' evidence of the future. I argue that this is the case with *mau* ('about to'/'want') in Indonesian, which is ambiguous between a proximate future (future *mau*) and a desire verb (desire *mau*) (Copley 2002, 2010, Tsilia to appear). I will show that future *mau* behaves like a direct evidential in presupposing that there is 'direct enough' evidence of its prejacent. Example (2) below illustrates how future *mau* conveys that rain is imminent, provided that there is direct evidence for it, such as dark clouds:

(2) Context: *There are dark clouds outside.*
(Ini) mau hujan.
It FUT rain
'It is about to rain.'

Future *mau* will thus fill this apparent typological gap, showing that the future is not incompatible with direct evidentials. As expected, the future is compatible with indirect evidentials. I will argue that the future marker *pasti* has a usage as an indirect evidential, akin to epistemic futures in Greek and Italian (see a.o., Giannakidou and Mari 2018, Ippolito and Farkas 2021, Frana and Menéndez-Benito 2019, 2023) and epistemic *must* (see a.o., von Fintel and Gillies 2010, 2021, Wurm 2023). I will propose an analysis of what counts as 'direct enough' evidence of the future. The latter is what future *mau*'s prejacent should be, and conversely, what *pasti*'s prejacent cannot be. Overall, my goal is to show that language has precisely a way of marking what counts as 'direct enough' evidence of the future, and that this is a linguistically meaningful category we should be describing.

2. Future marking in Colloquial Jakartan Indonesian

I start with some background on tense marking in Colloquial Jakartan Indonesian.[1] Indonesian in general does not have obligatory tense marking for the present or the past. Sentences are ambiguous between a present and a past tense interpretation, and the context disambiguates. Example (3) illustrates how the same utterance could be interpreted as a present or a past tense sentence, depending on the context.

(3) Lia nyanyi.
Lia sing
'Lia is signing/sang.' Tsilia (to appear):ex.1

[1] All Indonesian data come from original fieldwork with a native speaker of Colloquial Jakartan Indonesian, unless stated otherwise.

However, sentences are not three-way ambiguous between present, past, and future. To get a future interpretation, we need a future marker or a temporal phrase referring to the future, even if the context clearly disambiguates:

(4) Context: *What will Lia do at the event?*
Lia *(bakal) nyanyi.
Lia FUT sing
'Lia will sing.' Tsilia (to appear):ex.3

(5) Context: *I am finishing my BA, and someone asked me what I plan to do next.*
Aku (bakal) mulai kuliah magister bulan Mei.
I FUT start studying master in May
'I will start my Masters in May.'

Copley (2010) has described three future markers for Standard Indonesian, *akan*, *mau* (also 'want'), and *pasti*:

(6) Budi akan/mau/pasti makan ikan.
Budi FUT eat fish
'Budi will eat fish.' Standard Indonesian, Copley (2010):ex.1

I should note that there is a great deal of linguistic diversity in Indonesia, and that my consultant has grown up in the capital, Jakarta. Colloquial Jakartan Indonesian has the following three future markers: *bakal*, *mau* (also 'want'), and *pasti*. According to my consultant, *akan* is too formal in her dialect, and she instead uses *bakal* as the default future marker. To my knowledge, the only difference between *akan* and *bakal* is register.

In this paper, I will focus on the usage of *mau* as a proximate future, arguing that it is an evidential future presupposing that the attitude holder has 'direct enough' evidence of its prejacent. In other words, to use future *mau* the speaker needs to have observed a precursor of the future event, which is imminent. Despite the fact that, as we saw earlier, evidential constructions are usually limited to the present or past, future *mau* will serve as a proof of concept that natural language has a way of encoding what counts as 'direct enough' evidence of the future. I will then describe a usage of the future marker *pasti* as an indirect evidential. If *mau* indicates directness of its prejacent, *pasti* indicates indirectness when it is not used as a future marker (akin to epistemic *must*).

3. ***Mau* as a direct evidential future**

Mau in Indonesian can mean 'want' or 'about to'. In fact, there are two desire verbs, *pengen* and *mau*. When the meaning of 'want' is targeted, they can be used interchangeably, with no noticeable difference in their interpretations:

(7) Context: *I am finishing my BA, and someone asked me what I plan to do next.*

Aku mau/pengen kuliah program magister, tapi aku nggak bakal.
I WANT start program master but I NEG FUT
'I want to start a Masters program, but I will not.'

However, *mau* also has a life as a future marker, what I call future *mau* (as opposed to desire *mau* in Example (7)). We can detect future *mau* since it does not convey a desire meaning, does not require an animate attitude holder, and cannot be substituted by *pengen*. It also suffices as a future marker.[2] Future *mau* is a proximate future marker and a direct evidential, as seen in Example (8) below, where there is perceptual evidence of the train's imminent arrival.

(8) Context: *I'm in the train station and I hear the train whistle.*
Kereta=nya mau/#pengen datang.
train-DEF FUT come
'The train is about to arrive.'

Future *mau* asserts that its prejacent will occur in the near future, while also presupposing that the attitude holder has 'direct enough' evidence for it. Example (9) below further illustrates this, since the fall of the book is imminent, and future *mau* is licensed since there is visual evidence for the future event, namely the book being at the edge of the bookshelf.

(9) Context: *The book is at the edge of the bookshelf, I am afraid it will fall.*
Buku=nya mau jatuh buku=nya.
book.DEF FUT fall
'The book is about to fall.' Tsilia (to appear):ex.8

Future *mau* is both a future and a direct evidential, being what I call a *direct evidential future*, a proximate future presupposing that the attitude holder has firsthand evidence of a proposition which can be used to directly infer the prejacent. This is a category between direct and inferential evidentiality. The available evidence is not as direct as it gets, since it is not evidence for the prejacent itself, but rather of something leading to it. However, it is not indirect enough to be considered inferred evidence either. The evidence we have for the future event is as close as we can get to it; so close, that for the purposes of linguistic marking it is considered direct evidence of the future. In Example (9), seeing the book right at the edge of the bookshelf feels almost like watching it fall already. The intuition of my consultant is that there need to be "clear signs" for the future event.

Future *mau* differs from the regular future marker *bakal* in being a proximate future, as well as in requiring evidence for the future event.[3] However, not any kind of evidence will suffice; future *mau* is only licensed if the evidence-holder has firsthand evidence of a

[2]I will be glossing future *mau* as FUT and desire *mau* (also spelled out as *pengen*) as WANT.

[3]Further evidence that it behaves like an evidential comes from the fact that it cannot be interpreted in the scope of negation. A negated *mau* (*nggak mau*) can only receive a 'not want' interpretation, not a 'will not' one (Tsilia to appear).

proposition which can lead to directly inferring *mau*'s prejacent. In Example (9) the book's imminent fall can be directly inferred from the visual evidence. By contrast, in Example (10) below, the book's fall cannot be directly inferred from the available evidence, and thus future *mau* is not licensed; the regular future marker *bakal* would be used instead.

(10) Context: *The book is well-secured in the middle of the bookshelf. However, a magician in the room who wants to show off his skills says:*
Buku=nya bakal/#mau jatuh.
book-DEF FUT fall
'The book will fall.' Tsilia (to appear):ex.34a

Similarly, in a pure guess scenario, future *mau* cannot be used, since the prediction is not evidence-based. For example, if we toss a fair coin and make a guess, future *mau* is not licensed and the regular future marker *bakal* would be used instead:

(11) Context: *We're tossing a fair coin. I bet it will land heads, so I say:*
Yang muncul bakal/#mau kepala.
REL.CL appear FUT head
'It's going to be heads.'

Future *mau* behaves as a direct evidential future, since there needs to be 'direct enough' evidence of its prejacent. In other words, the available evidence should lead us directly (through a single reasoning step (Wurm 2023)) to the prejacent.

4. *Pasti* as an indirect evidential future

As we discussed, Indonesian also has another future, *pasti* (Copley 2010).

(12) Pasti hujan.
PASTI rain
'It'll rain.' (Copley 2010):ex.35a

It is the third future marker. I argue that in addition to being a future marker, *pasti* lives a double life as an indirect evidential marker, and can be used non-predictively. In its usage as an indirect evidential, *pasti* is (almost) *mau*'s mirror image. The prejacent of *mau* can be directly inferred, while the prejacent of *pasti* has to be *in*directly inferred.

The behavior of *pasti* patterns with other evidential futures described so far in the literature, i.e., the Greek and the Italian future (Mari 2010, Giannakidou and Mari 2018, Frana and Menéndez-Benito 2019, Ippolito and Farkas 2021, Frana and Menéndez-Benito 2023). These are future tense morphemes which can have a non-predictive usage, behaving like an indirect evidential or epistemic *must*. Just like *pasti*, they are future markers or indirect evidentials. They need at least one of these functions, but not necessary both simultaneously. Example (13) shows an instance of the Italian future's non-predictive, epistemic usage:

(13) Giacomo ora starà mangiando.
Giacomo now be.FUT.3SG eat.GERUND
'Giacomo must be eating now.' Italian, (Giannakidou and Mari 2018):ex.11b

These non-predictive uses of the future have been analyzed in terms of epistemic modality (Giannakidou and Mari 2018) or indirect evidentiality (Frana and Menéndez-Benito 2019, 2023). In Indonesian, *pasti* can also be used non-predictively as an indirect evidential, following the pattern of other evidential futures described so far.

(14) Context: *I'm in the room next to the kitchen, and all of a sudden I hear something shattering. I say to my partner:*
Pasti adanya pecah.
PASTI something broken
'Something must have broken.'

In Example (14), the prejacent of *pasti* is no longer a prediction; we have indirect evidence that it just occurred. *Pasti* as an indirect evidential is not limited to the past. Example (15) is about the present, and *pasti*'s prejacent is again indirectly inferred from the available evidence.

(15) Context: *I'm in Boston and I'm talking to someone in NY over the phone. They infer from my background sound that it's raining in Boston.*
Ini pasti lagi hujan.
it PASTI PROGR rain
'It must be raining.'

The evidence is indirect not because it is heard as opposed to seen—since Example (8) shows that the same type of evidence can count as 'direct' evidence for future *mau*. Crucially, what differentiates future *mau* and *pasti* is not the means through which the evidence is collected, but rather whether the evidence directly points to the prejacent. In other words, what matters in this evidential distinction is not so much the source of the evidence, but rather the way in which it leads to the prejacent (directly or indirectly).

5. Analysis: cognitive complexity matters

I analyze future *mau* and *pasti* drawing on ideas from analyses of epistemic *must* as an indirect evidential.[4] von Fintel and Gillies (2010, 2021) provide an account of epistemic *must* as an indirect evidential, according to which the evidence for the prejacent of *must* cannot be 'direct enough'. The main idea is that *must* is a necessity operator, and is thus strong, its perceived weakness being the result of its nature as an indirect evidential. In

[4] I do not attempt to summarize the extensive literature on epistemic *must* here, but I refer the interested reader to several works (see e.g., Stone (1994), von Fintel and Gillies (2010, 2021), Lassiter (2016), Goodhue (2017), Mandelkern (2019), Giannakidou and Mari (2016), Waldon (2021), Wurm (2023) a.o.)

Example (16b) there is direct evidence for the prejacent of *must*, and thus *must* is not licensed. By contrast, in Example (17b), there is no direct evidence for its prejacent.

(16) Context: *Seeing the pouring rain*

a. It's raining.
b. ??It must be raining. von Fintel and Gillies (2010):ex.6

(17) Context: *Billy seeing wet rain gear and knowing rain is the only cause*

a. It's raining.
b. It must be raining. von Fintel and Gillies (2010):ex.7

They formally implement this using the notion of a Kernel, a set of propositions describing what is 'direct enough'. Epistemic *must* requires that its prejacent is not settled (i.e., entailed or contradicted) by the Kernel.

Wurm (2023) amends this account of *must* as an indirect evidential, arguing that a binary distinction between what is direct enough and what is not in a context does not suffice to explain the behavior of epistemic *must*. She shows that the extent to which an inference is considered (in)direct depends on the speaker, the assumptions in the context, and essentially how much effort is required to infer the prejacent of epistemic *must*. Example (18), for instance, illustrates that directly perceiving *must*'s prejacent does not automatically disallow its use, as long as substantial reasoning effort is required to infer the prejacent. Conversely, Example (19) shows that indirect evidence might count as 'direct' in certain contexts, if the prejacent follows easily.

(18) Context: *A musicologist listening to a composition, noticing that the symphonic qualities can only be associated with Mozart:*
That must be Mozart. Wurm (2023):ex.7

(19) Context: *I am a therapist working in a group practice. As standard procedure, we put a red sign on our door handles during sessions. A new colleague asks whether she can print something in another colleague's room. I see a red sign:*
??Nein, die muss gerade Stunde haben.
no she must currently lesson have
'No, she must be in a session.' Wurm (2023):ex.9

She argues that language encodes reasoning effort. *Must* requires that its prejacent is inferred with substantial reasoning effort. She proposes a system encoding the number of reasoning steps required to reach the prejacent. If the number is below a contextually determined threshold, the prejacent is considered direct enough; otherwise it's indirect.

Indonesian corroborates the idea that natural language encodes cognitive effort, since future *mau* requires too little and *pasti* requires substantial cognitive effort to reach their prejacents. Inspired by Wurm's 2023 analysis of epistemic *must*, I argue that not only cognitive *complexity*, but also *simplicity* matters. In other words, Indonesian provides further evidence for her account. Epistemic *must* and *pasti* show us that certain modals require substantial reasoning effort to infer their prejacent. By contrast, future *mau* proves that minimal reasoning effort is not considered 'reasoning' for the purposes of language; instead, the inference to the prejacent of future *mau* is treated on a par with direct inferences.

For reasons of space, I can only provide a sketch of the formal analysis here. What needs to be formalized is what counts as "directly" arriving to the prejacent of a modal. The most "direct" way there is is having evidence for the prejacent itself. However, since the prejacent in the case of *mau* is in the future, and since we cannot directly perceive the future, we cannot directly perceive *mau*'s prejacent. For this reason, we cannot directly adopt a von Fintel and Gillies Kernel analysis, where there is only a set of propositions describing what is direct enough. This would work for epistemic *must* and *pasti*, but not for *mau*, since its prejacent will not be in the Kernel. What counts as direct *enough* for the future is directly perceiving something which given our prior beliefs can directly lead to the prejacent. For example, if we directly perceive dark clouds, and we have a prior belief that "if we perceive dark clouds, rain is imminent", then we can "directly", i.e., via one reasoning step, arrive to the prejacent "rain is imminent". Future *mau* will require at most one inferential step to its prejacent, while *pasti* at least two.

Formally, we can start from the notion of a Kernel (a set of propositions describing what is direct enough) from von Fintel and Gillies (2010, 2021), and then define a nested sequence of sets of propositions, arguing that all propositions in the first two sets are considered direct enough for the purposes of modals. Following Yalcin (2018), Wurm (2023), I utilize the notion of an awareness state, which for our purposes is equivalent to a Kernel. In other words, this is the set of propositions describing what is direct enough for the speaker in the context c_s at a world w in time t.

(20) Awareness state $\textsc{awa}_0(c_s,w,t)$ for c_s in w at t = $\{p : c_s$ believes and is aware of p in w at $t\}$.

This set of propositions describes what the speaker believes and is aware of, so it will include anything that the speaker directly observes, as well as any beliefs that are considered direct enough. To be able to distinguish between the prejacent of *pasti* and *mau*, we will define a nested sequence of awareness states, i.e., nested sets of sets of propositions. The prejacent of *must* or *pasti* cannot be in the first two awareness states, while the prejacent of *mau* has to be. We start by $\textsc{awa}_0(c_s,w,t)$, as defined above. This is the epicenter of the nested sets, and is equivalent to a Kernel (von Fintel and Gillies 2010, 2021). Then, we inductively define a nested sequence of awareness states. Each new awareness state includes the propositions of the previous one as well as any propositions that can be derived by two propositions in the previous awareness state. More formally:

(21) a. Base step:
$\textsc{awa}_1(c_s,w,t) = \textsc{awa}_0(c_s,w,t) \cup \{r : \exists p \exists q \in \textsc{awa}_0(c_s,w,t) : p \wedge q \subseteq r\}$

b. Inductive step:
For all $\kappa > 1$ in $\textsc{awa}_\kappa$, $\textsc{awa}_\kappa(c_s,w,t) = \textsc{awa}_{\kappa-1}(c_s,w,t) \cup$
$\{r : \exists p \exists q \in \textsc{awa}_{\kappa-1}(c_s,w,t) : p \wedge q \subseteq r\}$

We are thus left with a nested sequence of awareness states. Following von Fintel and Gillies (2010), an awareness state/Kernel settles a proposition if it entails or contradicts it. Given the notion of settling a proposition and the nested sequence of awareness states defined above, we can model what it means to have direct enough or sufficiently indirect evidence of the prejacent.

(22) a. If the prejacent is in $\text{AWA}_{\kappa}(c_s, w, t)$, where $\kappa \leq 1$, then the evidence is considered direct enough.
b. If the prejacent is in $\kappa \geq 2$ but not in $\kappa \leq 1$, then the evidence is considered sufficiently indirect.

The prejacent of future *mau* includes a prospective (proximate) aspectual marker responsible for the future meaning, and a proposition *p*. The prejacent of *pasti* includes only a proposition *p*. What *mau* and *pasti* have in common is that their prejacent is not in AWA_0. Where they differ is how many updates it takes to reach an awareness state, AWA_{κ}, which includes their prejacent ($\kappa \geq 2$ for *pasti* and $\kappa \leq 1$ for *mau*).[5]

6. Conclusion

Evidentials mark different degrees of robustness of evidence for a proposition. Our naive metaphysics instruct us that we cannot have direct evidence for a proposition about the future, thus one might expect that direct evidentials will not combine with future tenses. This expectation could apparently explain a typological gap or artificially create one. I argued that we should not let our naive metaphysics influence our linguistic analysis; although it is true that we cannot have direct evidence of the future, language has a way of talking about what counts as direct enough evidence of the future. Through a case study of two future modals in Colloquial Jakartan Indonesian, *mau* and *pasti*, I argued that natural language is sensitive to the simplicity or complexity of the inference from the available evidence to a modal's prejacent. Using ideas from the literature on epistemic *must* as an indirect evidential, I suggested that not only the prejacent itself is considered 'direct' evidence, but also propositions directly leading to it. So, a future tense can be a direct evidential, if the available evidence leads to its prejacent through one inference step. This serves as proof of concept that evidential marking on the future is not restricted to indirect evidentials. Language can encode what counts as 'direct' evidence of the future.

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[5]For reasons of space, I cannot I provide the Logical Forms to show the system in its full form, although I hope I conveyed the main idea.

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